5 Find the gradient of the curve with equation $y = f(x)$ at the point A where:

a $f(x) = x(x + 1)$ and A is at $(0, 0)$

b $f(x) = \dfrac{2x - 6}{x^2}$ and A is at $(3, 0)$

c $f(x) = \dfrac{1}{\sqrt{x}}$ and A is at $\left(\frac{1}{4}, 2\right)$

d $f(x) = 3x - \dfrac{4}{x^2}$ and A is at $(2, 5)$

E/P **6** $f(x) = \dfrac{12}{p\sqrt{x}} + x$, where p is a real constant and $x > 0$.

Given that $f'(2) = 3$, find p, giving your answer in the form $a\sqrt{2}$ where a is a rational number. **(4 marks)**

P **7** $f(x) = (2 - x)^9$

a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $f(x)$, giving each term in its simplest form.

b If x is small, so that x^2 and higher powers can be ignored, show that $f'(x) \approx 9216x - 2304$.

Hint Use the binomial expansion with $a = 2$, $b = -x$ and $n = 9$. ← Section 8.3

12.6 Gradients, tangents and normals

You can use the derivative to find the equation of the tangent to a curve at a given point. On the curve with equation $y = f(x)$, the gradient of the tangent at a point A with x-coordinate a will be $f'(a)$.

- **The tangent to the curve $y = f(x)$ at the point with coordinates $(a, f(a))$ has equation**

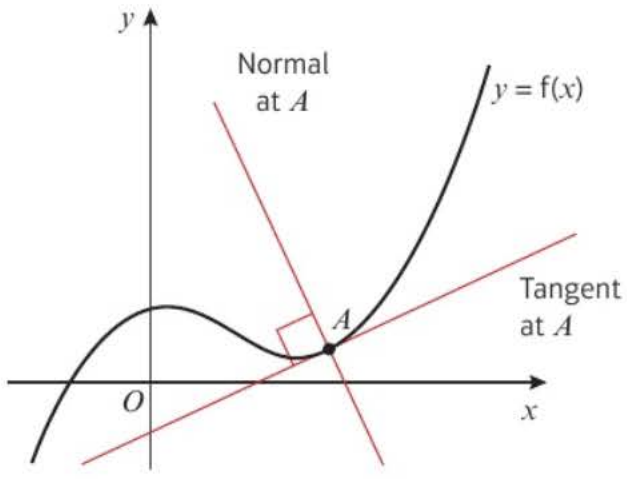
$$y - f(a) = f'(a)(x - a)$$

Links The equation of a straight line with gradient m that passes through the point (x_1, y_1) is $y - y_1 = m(x - x_1)$. ← Section 5.2

The **normal** to a curve at point A is the straight line through A which is perpendicular to the tangent to the curve at A. The gradient of the normal will be $-\dfrac{1}{f'(a)}$

- **The normal to the curve $y = f(x)$ at the point with coordinates $(a, f(a))$ has equation**

$$y - f(a) = -\frac{1}{f'(a)}(x - a)$$



Example 10

Find the equation of the tangent to the curve $y = x^3 - 3x^2 + 2x - 1$ at the point (3, 5).

$y = x^3 - 3x^2 + 2x - 1$

$\frac{dy}{dx} = 3x^2 - 6x + 2$

When $x = 3$, the gradient is 11.

So the equation of the tangent at (3, 5) is

$y - 5 = 11(x - 3)$

$y = 11x - 28$

First differentiate to determine the gradient function.

Then substitute for x to calculate the value of the gradient of the curve and of the tangent when $x = 3$.

You can now use the line equation and simplify.

Example 11

Find the equation of the normal to the curve with equation $y = 8 - 3\sqrt{x}$ at the point where $x = 4$.

$y = 8 - 3\sqrt{x}$

$= 8 - 3x^{\frac{1}{2}}$

$\frac{dy}{dx} = -\frac{3}{2}x^{-\frac{1}{2}}$

When $x = 4$, $y = 2$ and gradient of curve and of tangent $= -\frac{3}{4}$

So gradient of normal is $\frac{4}{3}$.

Equation of normal is

$y - 2 = \frac{4}{3}(x - 4)$

$3y - 6 = 4x - 16$

$3y - 4x + 10 = 0$

Write each term in the form ax^n and differentiate to obtain the gradient function, which you can use to find the gradient at any point.

Find the y-coordinate when $x = 4$ by substituting into the equation of the curve and calculating $8 - 3\sqrt{4} = 8 - 6 = 2$.

Find the gradient of the curve, by calculating

$$\frac{dy}{dx} = -\frac{3}{2}(4)^{-\frac{1}{2}} = -\frac{3}{2} \times \frac{1}{2} = -\frac{3}{4}$$

Gradient of normal $= -\dfrac{1}{\text{gradient of curve}}$

$$= -\frac{1}{\left(-\frac{3}{4}\right)} = \frac{4}{3}$$

Simplify by multiplying both sides by 3 and collecting terms.

Online Explore the tangent and normal to the curve using GeoGebra.

Exercise 12F

1 Find the equation of the tangent to the curve:

a $y = x^2 - 7x + 10$ at the point (2, 0)

b $y = x + \frac{1}{x}$ at the point $\left(2, 2\frac{1}{2}\right)$

c $y = 4\sqrt{x}$ at the point (9, 12)

d $y = \frac{2x - 1}{x}$ at the point (1, 1)

e $y = 2x^3 + 6x + 10$ at the point (−1, 2)

f $y = x^2 - \frac{7}{x^2}$ at the point (1, −6)

2 Find the equation of the normal to the curve:

a $y = x^2 - 5x$ at the point (6, 6)

b $y = x^2 - \frac{8}{\sqrt{x}}$ at the point (4, 12)

P **3** Find the coordinates of the point where the tangent to the curve $y = x^2 + 1$ at the point (2, 5) meets the normal to the same curve at the point (1, 2).

(P) **4** Find the equations of the normals to the curve $y = x + x^3$ at the points (0, 0) and (1, 2), and find the coordinates of the point where these normals meet.

(P) **5** For $f(x) = 12 - 4x + 2x^2$, find the equations of the tangent and the normal at the point where $x = -1$ on the curve with equation $y = f(x)$.

(E/P) **6** The point P with x-coordinate $\frac{1}{2}$ lies on the curve with equation $y = 2x^2$.
The normal to the curve at P intersects the curve at points P and Q.
Find the coordinates of Q. **(6 marks)**

Problem-solving

Draw a sketch showing the curve, the point P and the normal. This will help you check that your answer makes sense.

Challenge

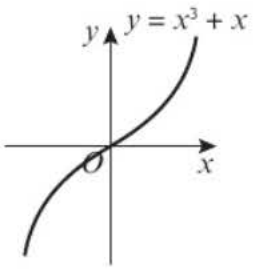
The line L is a tangent to the curve with equation $y = 4x^2 + 1$. L cuts the y-axis at $(0, -8)$ and has a positive gradient. Find the equation of L in the form $y = mx + c$.

Hint Use the discriminant to find the value of m when the line just touches the curve. ← Section 2.5

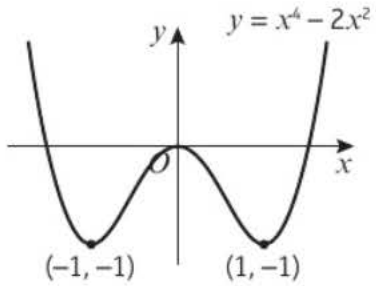
12.7 Increasing and decreasing functions

You can use the derivative to determine whether a function is **increasing** or **decreasing** on a given interval.

- **The function $f(x)$ is increasing on the interval $[a, b]$ if $f'(x) \geqslant 0$ for all values of x such that $a < x < b$.**
- **The function $f(x)$ is decreasing on the interval $[a, b]$ if $f'(x) \leqslant 0$ for all values of x such that $a < x < b$.**



The function $f(x) = x^3 + x$ is increasing for all real values of x.



The function $f(x) = x^4 - 2x^2$ is increasing on the interval $[-1, 0]$ and decreasing on the interval $[0, 1]$.

Notation The **interval** $[a, b]$ is the set of all real numbers, x, that satisfy $a \leqslant x \leqslant b$.

Example 12

Show that the function $f(x) = x^3 + 24x + 3$ is increasing for all real values of x.

$f(x) = x^3 + 24x + 3$
$f'(x) = 3x^2 + 24$
$x^2 \geqslant 0$ for all real values of x
So $3x^2 + 24 \geqslant 0$ for all real values of x.
So $f(x)$ is increasing for all real values of x.

First differentiate to obtain the gradient function.

State that the condition for an increasing function is met. In fact $f'(x) \geqslant 24$ for all real values of x.